

**LAPORAN PRAKTIKUM  
PRAKTIKUM PEMROGRAMAN OBJECT 1**

**MODUL 9  
PEWARISAN, POLIMORFISME DAN INTERFACE**

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# **BAB I**

## **HASIL PRAKTIKUM**

### **I.1 Program 1**

#### **I.1.A. Source Code**

```
public class Person {  
    protected String name;  
    protected String address;  
  
    public Person() {  
        System.out.println("Inside Person:Constructor");  
        name = "";  
        address = "";  
    }  
  
    public Person(String name, String address) {  
        this.name = name;  
        this.address = address;  
    }  
  
    public String getName() {  
        return name;  
    }  
  
    public String getAddress() {  
        return address;  
    }  
  
    public void setName(String name) {  
        this.name = name;  
    }  
    public void setAddress(String address) {  
        this.address = address;  
    }  
}
```

### I.1.B. Screenshot

```
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>javac person.java
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>java Person
Error: Main method not found in class Person, please define the main method as:
    public static void main(String[] args)
or a JavaFX application class must extend javafx.application.Application
```

*Gambar 1 Hasil Program 1*

### I.1.C. Analisis

Pada source code ini merupakan sebuah class person yang didalamnya terdapat beberapa atribut, maka Program ini tidak menampilkan output

## I.2 Program 2

### 1.2.A. Source Code

```
public class Student extends Person {

    public Student() {
        //super( "SomeName", "SomeAddress");
        //super();
        //super.name = "name";
        System.out.println("Inside Student:Constructor");
    }

    public static void main(String[] args) {
        Student anna = new Student();
    }

}
```

### 1.2.B. Screenshot

```
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>javac Student.java
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>java Student
Inside Person:Constructor
Inside Student:Constructor
```

*Gambar 2 Hasil Program 2*

### 1.2.C. Analisis

Pada source code ini merupakan sebuah main yang didalamnya terdapat sebuah Public class student extends Person untuk menyambungkan class kemudian terdapat sebuah perintah print1n, dan hasil ouput nya adalah inside person dan inside student.

## I.3 Program 3

### 1.3.A. Source Code

```
public class Pakaian2 {
    private int ID = 0;
    private String keterangan = "-keterangan diperlukan-";
    private double harga = 0.0;
```

```

private int jmlStok = 0;
private static int UNIQUE_ID = 0;

public Pakaian2() {
    ID = UNIQUE_ID++;
}

public int getID() {
    return ID;
}

public void setKeterangan(String d) {
    keterangan = d;
}

public String getKeterangan() {
    return keterangan;
}

public void setHarga(double p) {
    harga = p;
}

public double getHarga() {
    return harga;
}

public void setJmlStok(int q) {
    jmlStok = q;
}

public int getJmlStok() {
    return jmlStok;
}
}

class Baju extends Pakaian2 {
    private char kodeWarna = 'U';

    public void tampilInformasiBaju() {
        System.out.println("ID Baju: " + getID());
        System.out.println("Keterangan: " +
getKeterangan());
        System.out.println("Kode Warna: " + kodeWarna);
        System.out.println("Harga baju: " + getHarga());
        System.out.println("Jumlah stok: " + getJmlStok());
    }
}

```

### 1.3.B. Screenshot

```
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>javac Pakaian2.java
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>java Pakaian2
Error: Main method not found in class Pakaian2, please define the main method as:
    public static void main(String[] args)
or a JavaFX application class must extend javafx.application.Application
```

*Gambar 3 Hasil Program 3*

### 1.3.C. Analisis

Pada Source code ini adalah suatu Superclass and Subclass, jadi code ini mempunyai 2 class didalam nya terdapat class yang isinya sebuah variable dan class ke 2 nya adalah sebuah print1n yang akan di print pada class pertama. Namun setelah code dirun atau dijalankan tidak menampilkan apa apa

## I.4 Program 4

### 1.4.A. Source Code

```
public class Person {
    protected String name;
    protected String address;

    /**
     * Default constructor
     */
    public Person() {
        System.out.println("Inside Person:Constructor");
        name = "";
        address = "";
    }

    /**
     * Constructor dengan dua parameter
     */
    public Person(String name, String address) {
        this.name = name;
        this.address = address;
    }

    /**
     * Method accessor
     */
    public String getName() {
        System.out.println("Person Name : " + name);
        return name;
    }

    public String getAddress() {
        return address;
    }
}
```



```

    }

    public void setName(String name) {
        this.name = name;
    }

    public void setAddress(String add) {
        this.address = add;
    }
}

public class Student extends Person {
    public Student() {
        //super( "SomeName", "SomeAddress");
        //super();
        //super.name = "name";
        System.out.println("Inside Student:Constructor");
    }

    public String getName() {
        System.out.println("Student Name : " + name);
        return name;
    }

    public static void main(String[] args) {
        Student anna = new Student();
    }
}

public class Employee extends Person {
    public String getName() {
        System.out.println("Employee Name:" + name);
        return name;
    }

    public static void main(String[] args) {
        Person ref;
        Student studentObject = new Student();
        Employee employeeObject = new Employee();

        ref = studentObject; // Person menunjuk kepada
object Student

        String temp = ref.getName(); // getName dari
Student class dipanggil
        System.out.println(temp);

        ref = employeeObject; // Person menunjuk kepada
object Employee

```

```

        temp = ref.getName(); // getName dari Employee
class dipanggil
        System.out.println(temp);
    }
}

```

#### 1.4.B. Screenshot

```

D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>javac Person.java
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>javac Student.java
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>javac Employee.java
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>java Person
Error: Main method not found in class Person, please define the main method as:
  public static void main(String[] args)
or a JavaFX application class must extend javafx.application.Application
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>java Student
Inside Person:Constructor
Inside Student:Constructor
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>java Employee
Inside Person:Constructor
Inside Student:Constructor
Inside Person:Constructor
Student Name :
Employee Name:

```

*Gambar 4 Hasil Program 4*

#### 1.4.C. Analisis

Pada source code ini mempunyai tiga kelas diantaranya Person, Student, dan Employee. Student dan Employee merupakan turunan dari kelas Person. Namun setelah saya run code tersebut error dikarenakan terdapat tiga class dalam satu file. Maka dari itu saya memisahkan ke 3 class tersebut dan jadilah hasil code diatas.

### I.5 Program 5

#### 1.5.A. Source Code

```

public class PrivateElevator2 {
    private boolean bukaPintu = false;
    private int lantaiSkrng = 1;
    private int berat = 0;
    private final int KAPASITAS = 1000;
    private final int LANTAI_ATAS = 5;
    private final int LANTAI_BAWAH = 1;

    public void buka() {
        bukaPintu = true;
    }

    public void tutup() {
        hitungKapasitas();
        if (berat <= KAPASITAS) {
            bukaPintu = false;
        }
    }
}

```

```

        } else {
            System.out.println("Elevator kelebihan beban");
            System.out.println("Pintu akan tetap terbuka
sampai seseorang keluar");
        }
    }

    private void hitungKapasitas() {
        berat = (int) (Math.random() * 1500);
        System.out.println("Berat: " + berat);
    }

    public void naik() {
        if (!bukaPintu) {
            if (lantaiSkrng < LANTAI_ATAS) {
                lantaiSkrng++;
                System.out.println(lantaiSkrng);
            } else {
                System.out.println("Sudah mencapai lantai
atas");
            }
        } else {
            System.out.println("Pintu masih terbuka");
        }
    }

    public void turun() {
        if (!bukaPintu) {
            if (lantaiSkrng > LANTAI_BAWAH) {
                lantaiSkrng--;
                System.out.println(lantaiSkrng);
            } else {
                System.out.println("Sudah mencapai lantai
bawah");
            }
        } else {
            System.out.println("Pintu masih terbuka");
        }
    }

    public void setLantai(int tujuan) {
        if ((tujuan >= LANTAI_BAWAH) && (tujuan <=
LANTAI_ATAS)) {
            while (lantaiSkrng != tujuan) {
                if (lantaiSkrng < tujuan) {
                    naik();
                } else {
                    turun();
                }
            }
        } else {

```

```

        System.out.println("Lantai Salah");
    }
}

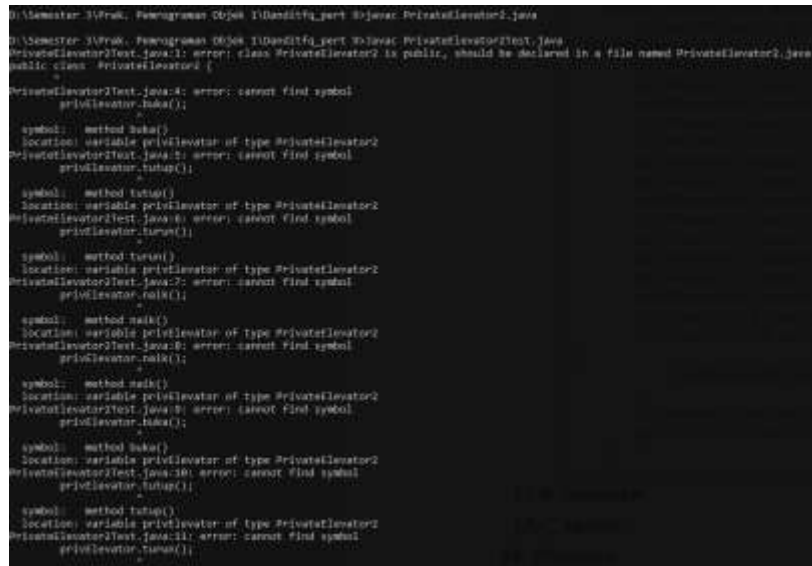
public int getLantai() {
    return lantaiSkrng;
}

public boolean getStatusPintu() {
    return bukaPintu;
}
}

public class PrivateElevator2Test {
    public static void main(String args[]) {
        PrivateElevator2 privElevator = new
PrivateElevator2();
        privElevator.buka();
        privElevator.tutup();
        privElevator.turun();
        privElevator.naik();
        privElevator.naik();
        privElevator.buka();
        privElevator.tutup();
        privElevator.turun();
        privElevator.buka();
        privElevator.turun();
        privElevator.tutup();
        privElevator.turun();
        privElevator.turun();
        int lantai = privElevator.getLantai();
        if (lantai != 5 && !privElevator.getStatusPintu())
        {
            privElevator.setLantai(5);
        }
        privElevator.setLantai(10);
        privElevator.buka();
    }
}

```

### 1.5.B. Screenshot



```
D:\Semester 3\Prak. Pemrograman Objek 1\DanDita_perk 3\Java2 PrivateElevator2.java
D:\Semester 3\Prak. Pemrograman Objek 1\DanDita_perk 3\Java2 PrivateElevator2Test.java
PrivateElevator2Test.java:1: error: class PrivateElevator2 is public, should be declared in a file named PrivateElevator2.java
public class PrivateElevator2 {
^
PrivateElevator2Test.java:4: error: cannot find symbol
    privateElevator.buka();
                    ^
    symbol:   method buka()
    location: variable privateElevator of type PrivateElevator2
PrivateElevator2Test.java:5: error: cannot find symbol
    privateElevator.tutup();
                    ^
    symbol:   method tutup()
    location: variable privateElevator of type PrivateElevator2
PrivateElevator2Test.java:6: error: cannot find symbol
    privateElevator.turun();
                    ^
    symbol:   method turun()
    location: variable privateElevator of type PrivateElevator2
PrivateElevator2Test.java:7: error: cannot find symbol
    privateElevator.naik();
                    ^
    symbol:   method naik()
    location: variable privateElevator of type PrivateElevator2
PrivateElevator2Test.java:8: error: cannot find symbol
    privateElevator.naik();
                    ^
    symbol:   method naik()
    location: variable privateElevator of type PrivateElevator2
PrivateElevator2Test.java:9: error: cannot find symbol
    privateElevator.buka();
                    ^
    symbol:   method buka()
    location: variable privateElevator of type PrivateElevator2
PrivateElevator2Test.java:10: error: cannot find symbol
    privateElevator.tutup();
                    ^
    symbol:   method tutup()
    location: variable privateElevator of type PrivateElevator2
PrivateElevator2Test.java:11: error: cannot find symbol
    privateElevator.turun();
                    ^
    symbol:   method turun()
    location: variable privateElevator of type PrivateElevator2
```

Gambar 5 Hasil Program 5

### 1.5.C. Analisis

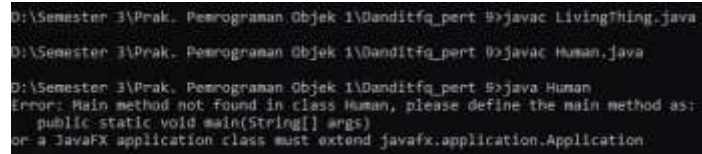
Pada source code ini mempunyai 2 nama file didalamnya yaitu PrivateElevator dan Testnya, setelah saya compile file ini tidak bisa dikarnakan adanya 2 class didalamnya, namun setelah saya pisah ke2 class tersebut terdapat 19 error dalam mainnya dikarnakan kurangnya atribut pada classnya.

## 1.6 Program 6

### 1.6.A. Source Code

```
public abstract class LivingThing {
    public void breath() {
        System.out.println("Living Thing breathing...");
    }
    public void eat() {
        System.out.println("Living Thing eating...");
    }
    /**
     * Abstract method walk
     * Kita ingin method ini di-overridden oleh subclasses
     */
    public abstract void walk();
}
public class Human extends LivingThing {
    public void walk() {
        System.out.println("Human walks...");
    }
}
```

### 1.6.B. Screenshot



```
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfa_pert 5>javac LivingThing.java
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfa_pert 5>javac Human.java
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfa_pert 5>java Human
Error: Main method not found in class Human, please define the main method as:
  public static void main(String[] args)
or a JavaFX application class must extend javafx.application.Application
```

Gambar 6 Hasil Program 6

### 1.6.C. Analisis

Pada source code ini didalamnya terdapat 2 class yang menyebabkan file code ini tidak bisa dirunning, namun setelah class ini dipisahkan code bisa di running

## I.7 Program 7

### 1.7.A. Source Code

```
public interface Relation {
    boolean isGreater(Object a, Object b);
    boolean isLess(Object a, Object b);
    boolean isEqual(Object a, Object b);
}

public class Line implements Relation {
    private double x1;
    private double x2;
    private double y1;
    private double y2;

    public Line(double x1, double x2, double y1, double y2)
    {
        this.x1 = x1;
        this.x2 = x2;
        this.y1 = y1;
        this.y2 = y2;
    }

    public double getLength() {
        double length = Math.sqrt((x2 - x1) * (x2 - x1) +
(y2 - y1) * (y2 - y1));
        return length;
    }

    @Override
    public boolean isGreater(Object a, Object b) {
        double aLen = ((Line) a).getLength();
        double bLen = ((Line) b).getLength();
        return (aLen > bLen);
    }
}
```

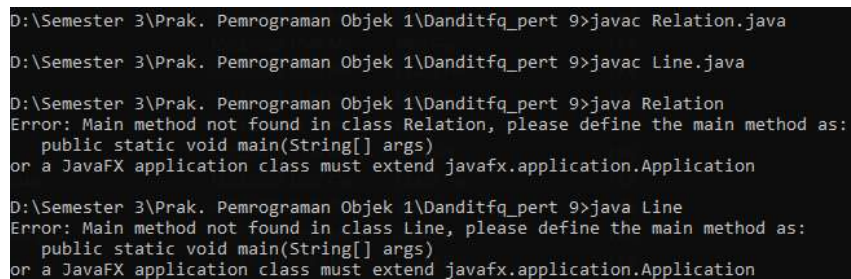
```

@Override
public boolean isLess(Object a, Object b) {
    double aLen = ((Line) a).getLength();
    double bLen = ((Line) b).getLength();
    return (aLen < bLen);
}

@Override
public boolean isEqual(Object a, Object b) {
    double aLen = ((Line) a).getLength();
    double bLen = ((Line) b).getLength();
    return (aLen == bLen);
}
}

```

### 1.7.B. Screenshot



```

D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>javac Relation.java
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>javac Line.java
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>java Relation
Error: Main method not found in class Relation, please define the main method as:
  public static void main(String[] args)
or a JavaFX application class must extend javafx.application.Application
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>java Line
Error: Main method not found in class Line, please define the main method as:
  public static void main(String[] args)
or a JavaFX application class must extend javafx.application.Application

```

*Gambar 7 Hasil Tugas 7*

### 1.7.C. Analisis

Pada source code ini terdapat class yang didalamnya terdapat beberapa object dan terdapat override. Code ini tidak bisa dijalankan dikarenakan kurangnya main.

## BAB II

### TUGAS PRAKTIKUM

#### II.1. Tugas Program 1-2

##### II.1.A. Source Code

###### a. Tugas 1

```
public class ComputerScienceStudent extends StudentRecord {
    private String[] programmingLanguages;

    public ComputerScienceStudent() {
        super();
    }

    public String[] getProgrammingLanguages() {
        return programmingLanguages;
    }

    public void setProgrammingLanguages(String[]
programmingLanguages) {
        this.programmingLanguages = programmingLanguages;
    }

    // Override metode print dari superclass

    public void print() {
        super.print(""); // Hapus argumen temp yang tidak
diperlukan
        if (programmingLanguages != null) {
            System.out.println("Programming Languages: " +
String.join(", ", programmingLanguages));
        }
    }

    // Metode baru yang spesifik untuk mahasiswa komputer
    public void code() {
        System.out.println(getName() + " is coding.");
    }

    // Contoh penggunaan
    public static void main(String[] args) {
        ComputerScienceStudent computerStudent = new
ComputerScienceStudent();
        computerStudent.setName("John");
        computerStudent.setAge(20);
        computerStudent.setEnglishGrade(85);
        computerStudent.setMathGrade(90);
        computerStudent.setScienceGrade(88);
    }
}
```



```

        computerStudent.setProgrammingLanguages(new
String[]{"Java", "Python", "C++"});

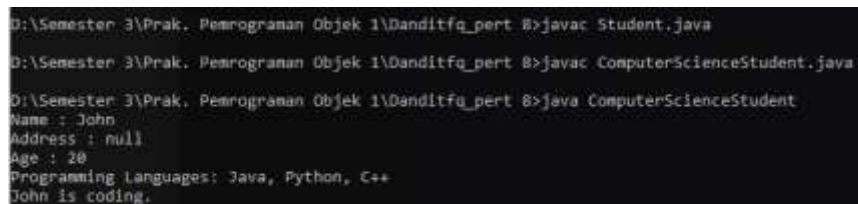
        // Memanggil metode print dari subkelas
computerStudent.print();

        // Memanggil metode code dari subkelas
computerStudent.code();
    }
}

```

## b. Tugas 2

### II.1.B. Screenshot

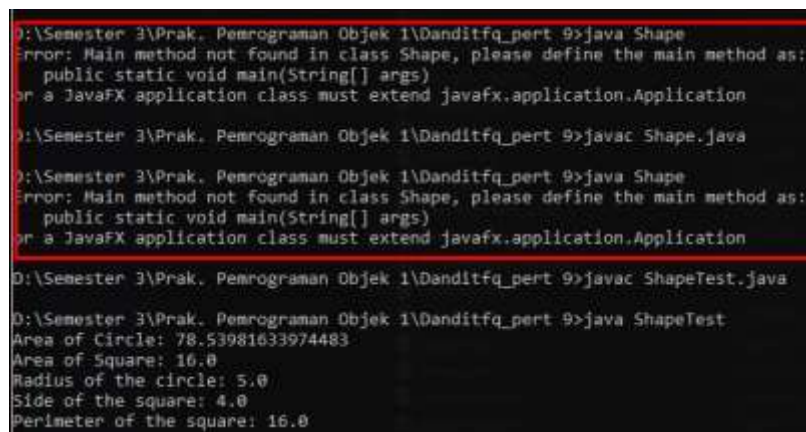


```

D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 8>javac Student.java
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 8>javac ComputerScienceStudent.java
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 8>java ComputerScienceStudent
Name : John
Address : null
Age : 20
Programming Languages: Java, Python, C++
John is coding.

```

Gambar 8 Hasil Tugas 1



```

D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>java Shape
Error: Main method not found in class Shape, please define the main method as:
public static void main(String[] args)
or a JavaFX application class must extend javafx.application.Application
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>javac Shape.java
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>java Shape
Error: Main method not found in class Shape, please define the main method as:
public static void main(String[] args)
or a JavaFX application class must extend javafx.application.Application
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>javac ShapeTest.java
D:\Semester 3\Prak. Pemrograman Objek 1\Danditfq_pert 9>java ShapeTest
Area of Circle: 78.53981633974483
Area of Square: 16.0
Radius of the circle: 5.0
Side of the square: 4.0
Perimeter of the square: 16.0

```

Gambar 9 Hasil Tugas 2

### II.1.C. Analisa

1. Pada source code tugas 1 membuat subkelas dari StudentRecord yang telah dibuat sebelumnya pada modul 8, pada class ini menambahkan informasi bahasa komputer yang dipelajari , serta menambahkan beberapa metode, dan hasilnya pada *Gambar 8 Hasil Tugas 1*.
2. Pada source code tugas 2 ini membuat kelas abstrak Shape, dengan method abstrak `getArea()` dan `getName()`, kemudian dibuat dua subclasses yaitu Circle dan Square. Dan hasilnya pada *Gambar 9 Hasil Tugas 2*.

### **BAB III**

### **KESIMPULAN**

Pada pertemuan ke 9 ini saya mempelajari dari Modul 9 yang berjudul “Pewarisan, polimorfisme dan interface”.Pewarisan memberikan cara untuk membuat hierarki kelas dan membagikan sifat dan perilaku antara kelas-kelas terkait, untuk polimorfisme memungkinkan menangani objek dari berbagai kelas dengan cara yang seragam, meningkatkan fleksibilitas dan reusabilitas, dan interface memberikan cara untuk mendefinisikan kontrak yang harus dipenuhi oleh kelas-kelas yang berbeda, memisahkan antarmuka pengguna dengan implementasi, dan mendukung polimorfisme.